VISUAL BILL OF MATERIALS (BOM)

IoT Patient Health & Safety Monitor

Component visual reference, hardware specifications, interfaces, and procurement budget

ESTIMATED BUDGET

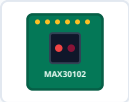
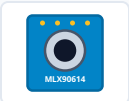
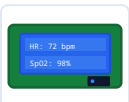
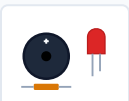
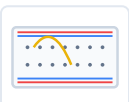
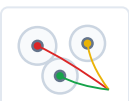
₹3,200

11 Core Components & Sensor Packages

ARCHITECTURE

ESP32 Dual-Core (Wi-Fi + BLE)

I²C Bus (4 devices) + UART GPS + Analog ECG

VISUAL	COMPONENT	FUNCTION	INTERFACE	KEY TECHNICAL SPECIFICATIONS	COST
	ESP32 DevKit 30/38-pin NodeMCU	MCU & Wireless Gateway	`Wi-Fi + BLE`	Xtensa dual-core LX6 240MHz, 3.3V logic, 5V input via USB	**₹350**
	**MAX30102** Optical Oximeter	Pulse Rate & SpO_2 Sensor	`I²C (0x57)`	Red (660nm) & IR (880nm) LEDs, 18-bit ADC, 3.3V rail	**₹150**
	AD8232 Module 3-Lead ECG	Cardiac Electrical Activity	`Analog (GPIO 34)`	Gain=100 instrumentation amp, leads-off detect, 3.3V supply	**₹700**
	**MLX90614** Infrared Pyrometer	Non-Contact Thermometer	`I²C / SMBus (0x5A)`	±0.2°C medical zone accuracy, 2–5cm optimal distance	**₹600**
	MPU6050 6-DOF IMU	Fall & Impact Detection	`I²C (0x68)`	3-axis accelerometer + 3-axis gyro, onboard DMP engine	**₹150**
	NEO-6M GPS Ceramic Patch Ant	Location Tracking	`UART2 (TX/RX)`	50 channels, 9600 baud rate, NMEA-0183 output protocol	**₹250**
	**16×2 LCD + I²C** PCF8574 Adapter	Patient Vital Display	`I²C (0x27 / 0x3F)`	HD44780 controller, 4 pins total, contrast trimmer, 5V input	**₹160**
	**Buzzer & LEDs** Audio/Visual Alert	Emergency Alert System	`Digital GPIO`	5V Active buzzer (~20mA), 5mm LEDs + 220Ω / 330Ω resistors	**₹50**
	**Breadboard + Wires** 830-Point + Jumpers	Prototyping Circuit	`0.1" Dupont`	Solderless MB-102 board, male-male & male-female jumper wires	**₹200**
	Battery + TP4056 Li-ion Charger & Cell	Portable Power Supply	`Micro-USB / Type-C`	3.7V 18650 Li-ion (2200–2600mAh) + DW01A protection circuit	**₹300**
	**Electrodes & Misc.** Gel Pads + Cable	ECG Accessories & Power	`3.5mm Jack`	3× Ag/AgCl disposable adhesive pads, 3-lead snap connector, switch	**₹300**
				ESTIMATED TOTAL COST:	**~₹3,210**

Bus Sharing & Interface Topology

- **I²C Shared Bus (GPIO 21 [SDA] & GPIO 22 [SCL]):** 4 devices share the same 2 wires: MAX30102 (`0x57`), MLX90614 (`0x5A`), MPU6050 (`0x68`), and 16×2 LCD Backpack (`0x27/0x3F`).

- **UART2 Serial (GPIO 16 [RX2] & GPIO 17 [TX2]):** Connects to NEO-6M GPS TX/RX at 9600 baud.
- **Analog ADC1 (GPIO 34):** Connects directly to AD8232 OUTPUT pin to avoid ADC2 conflicts when Wi-Fi is active.
- **Power Rails:** Feed 5V (VIN) to 16×2 LCD for sharp contrast; feed clean 3.3V to AD8232, MAX30102, and MLX90614.